NAME: _____ NetID: _____

INFO 4220 – Networks II: Market Design
Spring 2025 – Prelim 2

Solution

Problem 1 (16 points)

1.1 Explain the difference between matching problems and assignment problems. What criteria are used in each to choose "good" matchings or assignments? (5 pts)

Answer:

In a matching problem, you match agents to agents, so both sides have preferences. In this type of problems, we look for solutions that are stable.

1 point for explaining that are agent to agent so both sides have preferences.
1.5 points for explaining that we look for solutions that are stable.

In an assignment problem, we match agents to objects. Agents have preferences over objects, but objects do not have preferences. Objects may have owners (endowments), or people may have priority to get certain object.
In this type of problems, we want assignments that are efficient.

1 point for explaining that we match agents to object (1 have preferences, the other does not).
1.5 points for explaining that we look for solutions that are stable.

1.2 Does a serial random dictatorship always produce assignments that are in the core for assignment problems with private endowments? If yes, explain why. If not, provide a counterexample. (5 pts)

Solution:

In assignment problems with private endowments, the random serial dictatorship does not produce individually rational assignments or assignments in the core, as some people may end up with objects than are worst than their initial endowment, or may better off by trading between themselves. ***2 points***

Consider the following example

Antonio (Endowment A)	*Bruno (Endowment B)*	*Camilo (Endowment C)*	*Dolores (Endowment D)*
B	*A*	*A*	*B*
C	*C*	*B*	*C*
A	*B*	*C*	*D*
D	*D*	*D*	*A*

In this example, if the order of dictators is Camilo – Dolores - Bruno – Antonio, the assignment would be Camilo – A; Dolores B; Bruno – C; Antonio - D. This assignment is not in the core, because Antonio and Bruno would be better off by not participating in the market and trading objects between them.
3 points.

1.3 Consider a course allocation problem with 3 students and 4 courses (Students' true preferences over courses are in the table below). Students can enroll in 2 courses each. Course 1 has a capacity of 1, course 2 has a capacity of 2, course 3 has a capacity of 3, and course 3 has a capacity of 4.

S_1	S_2	S_3
C_1	C_2	C_3
C_2	C_3	C_1
C_3	C_1	C_2
C_4	C_4	C_4

Assume that students bid in the following way in the Gale-Shapley Pareto Dominant Market Mechanism:

Capacity	1	2	3	3
Course	C_1	C_2	C_3	C_4
S_1	75	20	3	2
S_2	0	98	1	1
S_3	50	40	5	5

a. Find the course allocation using the preferences and bids above (show and explain each step). (2 pts)

Solution:

Step 1:
S1: C1, C2.
S2: C2, C3.
S3: C3, C1.

S1 gets C1 and C2.
S2 gets C2 and C3.
S3 gets C3, but is rejected in C1.

Step 2:
S3: C2.
C2 rejects S1 and goes to S3.

Step 3:
S1: C3

The allocation will be:
S1: C1, C3
S2: C2, C3
S3: C2, C3.

1 point for the solution, 1 point for showing the steps.

b. Will S_3 want to adjust their bids? If so, how? (2 pts)

Answer: Yes, he will bid higher for C1 (more than 75). ***2 points****.*

c. If S_3 does what you suggest in (b) (i.e. adjust their bid or not), will anyone else want to adjust their bid? (2 pts)

Answer: Yes, S1 will react and bid higher for C1. ***2 point.***

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Problem 2 (16 points, 4 pts each part)

In a school matching problem, there are 3 students (Antonio, Bruno, Camilo), and 3 schools (North, South, East). Each school has a capacity of 1, and preferences and priorities are as follows:

Antonio	**Bruno**	**Camilo**
South	South	East
East	North	South
North	East	North

North	**South**	**East**
Antonio	Camilo	Antonio
Bruno	Bruno	Camilo
Camilo	Antonio	Bruno

a) Find the school assignment that results from using deferred acceptance with students proposing

ANSWER:
Step 1:

Antonio – South (rejected)
Bruno – South
Camilo – East

Step 2:
Antonio – East
East rejects Camilo.

Step 3:
Camilo – South
South rejects Bruno

Step 4:
Bruno – North.

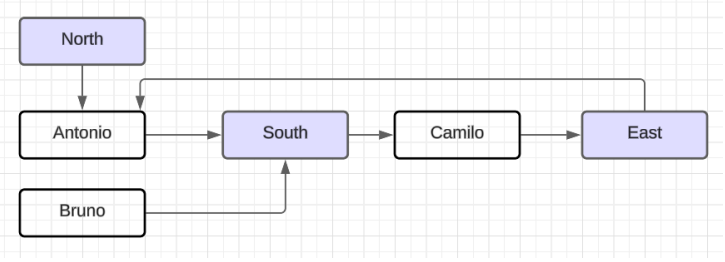
{Antonio – East, Bruno – North, Camilo – South}
2 points for the results, 2 points for the steps.

b) Is the matching you found in (a) Pareto efficient for students? Explain clearly why

*No **(1 point)**, Camilo and Antonio could switch and be better **(3 points)**.*

c) Find the school assignment that results from using top trading cycles

ANSWER:
Step 1:



Antonio – South
Camilo – East

Step 2

Bruno – North.

2 points for step 1
1 point for step 2
1 point for solution

d) Is the matching you found in (c) stable? Explain clearly why

*ANSER: No, it is not stable **(1 point)** because there is justified envy: Bruno prefers South to his assignment, and has a higher priority than Antonio at South. **(3 points)***

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Problem 3 (18 points)

Three individuals are trying to divide 3 cheescakes of different flavors: Plain, Chocolate, and Berries. Their preferences are as follows:

Amelia	**Benjamin**	**Charlotte**
Plain	Plain	Berries
Berries	Chocolate	Plain
Chocolate	Berries	Chocolate

a) Find the random assignment that results from using a random serial dictatorship (4 pts)

Order	*Amelia*	*Benjamin*	*Charlotte*
A – B – C	*P*	*C*	*B*
A – C – B	*P*	*C*	*B*
B – A – C	*B*	*P*	*C*
B – C – A	*C*	*P*	*B*
C – A – B	*P*	*C*	*B*
C – B - A	*C*	*P*	*B*

2 points

	P	*B*	*C*
A	*½*	*1/6*	*1/3*
B	*1/2*	*0*	*½*
C	*0*	*5/6*	*1/6*

2 points

b) Find the random assignment that results from using the eating algorithm if Amelia and Benjamin eat at the same speed (let's say they eat at a speed of 1), and Charlotte eats twice as fast (she eats at a speed of 2) (4 pts) Hint: Remember that no one can eat more than an entire object

ANSWER:
Step 1:
Amelia and Benjamin eat ½ of plain each.
Charlotte eats the entire berry.

Step 2:
Amelia and Benjamin eat ½ of Chocolate each.
2 points

	Plain	*Berry*	*Chocolate*
Amelia	*½*	*0*	*½*
Benjamin	*½*	*0*	*½*
Charlotte	*0*	*1*	*0*

2 points

c) Is the assignment you found in part (b) ex-post efficient? Explicitly show why or why not (5 pts)

ANSWER:

$$\frac{1}{2}\begin{pmatrix}1 & 0 & 0\\ 0 & 0 & 1\\ 0 & 1 & 0\end{pmatrix}+\frac{1}{2}\begin{pmatrix}0 & 0 & 1\\ 1 & 0 & 0\\ 0 & 1 & 0\end{pmatrix}$$

Alternative: Explain the 2 potential assignments with words instead of matrices.

Both are efficient, so the assignment is ex/post efficient.

3 points explanation.
2 points result.

d) Does the assignment you found in part (b) stochastically dominate the assignment you found in part (a). Explicitly show why or why not (5 pts)

ANSWER:

	SRD	*EA*
Amelia	*1st: ½* *1st or 2nd : 2/3*	*1st: ½* *2nd : ½*
Benjamin	*1st : ½* *1st or 2nd: 1*	*1st : ½* *1st or 2nd: 1*
Charlotte	*1st: 5/6* *1st or 2nd: 1*	*1st: 1*

It is not stochastically dominant, because for Amelia SRD is better than EA.

2 points for the result
3 points for the explanation.

Note: Showing that for Amelia SRD is better than EA is enough to get full credits.